Proc. Natl. Acad. Sci. USA
Vol. 83, pp. 53–57, January 1986
Biochemistry

An arachidonate metabolite is involved in the conversion from α_1- to β-adrenergic glycogenolysis in isolated rat liver cells

(α_1- and β-adrenoceptors/cyclooxygenase)

EDWARD J. N. ISHAC AND GEORGE KUNOS*

Departments of Pharmacology and Therapeutics and of Medicine, McGill University, Montreal, PQ H3G 1Y6, Canada

Communicated by Karl H. Beyer, Jr., August 30, 1985

ABSTRACT ***In vitro*** **incubation of isolated rat liver cells in a serum-free buffer leads to the suppression of the glycogenolytic effect of phenylephrine and the simultaneous emergence of a glycogenolytic response to isoproterenol within 4 hr. This time-dependent conversion of the adrenergic receptor response from α_1 to β type is prevented by the presence in the incubation medium of 0.5% fatty-acid-free, but not regular, bovine serum albumin. A 20-min exposure of freshly isolated liver cells to arachidonic acid (10 μg/ml), but not to stearic or palmitic acid, causes an acute shift in the receptor response from α_1 to mixed α_1/β type, similar in direction to that seen after prolonged incubation of the cells. This effect of arachidonic acid is prevented by 0.2 μM ibuprofen but not by the same concentration of nordihydroguaiaretic acid. Ibuprofen (1 μM) or indomethacin (1 μM) also inhibits the time-dependent shift in the receptor response. Actinomycin D inhibits the change in receptor response that is caused by prolonged incubation but not the change that is caused by exogenous arachidonic acid. It is proposed that the time-dependent conversion from α_1- to β-adrenergic receptor-mediated glycogenolysis in isolated rat liver cells is related to a parallel increase in the phospholipase-mediated release of arachidonic acid and the subsequent formation of a key cyclooxygenase metabolite. A protein factor appears to be involved in the regulation of the release of arachidonic acid but not in the action of its metabolite. A possible mechanism by which this metabolite may regulate inverse changes in the coupling of α_1- and β-receptors to postreceptor pathways is discussed.**

Adrenergic receptor mechanisms in the rat liver display unusual plasticity. In the liver of adult, male rats, the glycogenolytic effect of catecholamines is mediated by a calcium-linked, cAMP-independent, α_1-adrenergic mechanism (1). The same effect, however, is mediated by a predominantly β-adrenergic, cAMP-dependent mechanism in the liver of female (2) and fetal rats (3) and in male rats under certain conditions, including thyroid (4, 5) and glucocorticoid deficiency (6), malignant transformation (7), cholestasis (8), fasting (9), and liver regeneration (8, 10). A conversion from α_1- to predominantly β-adrenergic glycogenolysis has also been observed *in vitro*, during primary culturing of rat hepatocytes (11), or after short-term incubation of the isolated cells in a serum-free medium (12). By using the latter system, we have recently observed that lipomodulin, an endogenous, phospholipase-inhibitory protein (13), can reverse the time-dependent shift from α_1- to β-receptor mediated glycogenolysis, whereas melittin, an activator of phospholipase A_2, accelerates it (12). These findings have suggested that the time-dependent conversion of the hepatic adrenergic receptor response is related to a parallel increase in membrane phospholipase A_2 activity (12), but have not clarified the possible mechanism of this effect. Phospholipase A_2 catalyzes the release from membrane phospholipids of arachidonic acid, the precursor of eicosanoids. The present experiments provide evidence that an arachidonate metabolite, possibly generated through the cyclooxygenase pathway, is directly involved in the time-dependent conversion from α_1- to β-adrenergic glycogenolysis in isolated rat liver cells.

MATERIALS AND METHODS

Liver cells were isolated from male Sprague–Dawley rats weighing 180–250 g by a collagenase perfusion technique (4, 14). The cells were suspended in Krebs–Henseleit buffer containing (in mM) NaCl 115, KCl 3.7, $MgSO_4$ 1.2, KH_2PO_4 2.2, $NaHCO_3$ 25, glucose 40, as well as 1.5% gelatin (Difco) at pH 7.4. To minimize depletion of drugs and other substances added to the medium through uptake and metabolism by the cells (1), dilute cell suspensions were used throughout (4 mg of wet cells per ml of buffer). Certain batches of cells were incubated in a medium containing 0.5% bovine serum albumin (BSA), either regular or fatty-acid-free, as indicated. In such cases, the concentration of gelatin was reduced to 1%. Cell suspensions were incubated at 37°C under an atmosphere of 5% CO_2/95% O_2 and with continuous shaking at 100 cycles per min. All cells were preincubated for 30 min, necessary to achieve stable basal levels of phosphorylase *a* activity (4). After this, the effects of various drugs or other substances on phosphorylase *a* activity were tested either immediately (0-hr cells) or following a further 4 hr incubation (4-hr cells). Phosphorylase *a* activity was assayed as described (4, 13), by measuring the incorporation of [^{14}C]glucose 1-phosphate into glycogen. Units of enzyme activity are nmol of [^{14}C]glucose incorporated into glycogen per min per mg of protein at 31°C. Protein in the 3000 × *g* supernatant was measured by the method of Lowry *et al.* (15).

Materials. BSA, regular or fatty-acid-free, L-phenylephrine hydrochloride, L-isoproterenol hydrochloride, L-epinephrine bitartrate, arachidonic acid, stearic acid, palmitic acid, actinomycin D, ibuprofen, indomethacin, and nordihydroguaiaretic acid (NDGA) were from Sigma. *dl*-Propranolol hydrochloride was from Ayerst Laboratories (New York), and prazosin hydrochloride was a gift from Pfizer of Canada. α-D-[U-^{14}C]glucose 1-phosphate (200 mCi/mmol; 1 Ci = 37 GBq) was from New England Nuclear. Other chemicals were from usual commercial sources.

The publication costs of this article were defrayed in part by page charge payment. This article must therefore be hereby marked "*advertisement*" in accordance with 18 U.S.C. §1734 solely to indicate this fact.

Abbreviations: BSA, bovine serum albumin; NDGA, nordihydroguaiaretic acid.

*To whom reprint requests should be addressed at: Department of Pharmacology, McGill University, 3655 Drummond Street, Montreal, PQ H3G 1Y6, Canada.

RESULTS

In vitro incubation of isolated liver cells for 4 hr in a serum-free buffer results in the suppression of the phosphorylase-activating effect of the α-adrenergic agonist phenylephrine and the simultaneous emergence of a stimulatory response to the β-adrenergic agonist isoproterenol as reported earlier (see ref. 12, and Fig. 1*A* and *B*). The results of this earlier study indicate that this apparent conversion of the adrenergic receptor response from α_1 to β type may be related to increased activity of membrane phospholipase A_2 (12). This enzyme catalyzes the hydrolytic cleavage of fatty acids from the 2 position of membrane phosphatides, with the lysophosphatide residues accumulating in the cell membrane. Fatty-acid-free BSA avidly binds fatty acids, and it can be used as a lipid trap to continuously remove fatty acids from the medium. To test the possible involvement of fatty acids released by liver cells in the altered receptor response, fatty-acid-free BSA was added to the medium of an aliquot of cells. Fatty-acid-free BSA did not influence the adrenergic response of freshly isolated cells (Fig. 1). However, when present in the medium throughout the 4 hr incubation period, it completely prevented the change in receptor response: phenylephrine was as effective as in 0-hr cells, and isoproterenol remained inactive (Fig. 1*D*). Regular BSA, which is saturated with fatty acids, did not have a similar effect when used at the same concentration, as it did not prevent the suppression of the response to phenylephrine and the simultaneous emergence of a response to isoproterenol in the 4-hr cells (Fig. 1*C*). This indicates that the effects of fatty-acid-free BSA are due to the removal of fatty acids from the medium and not to an action of albumin itself. The finding

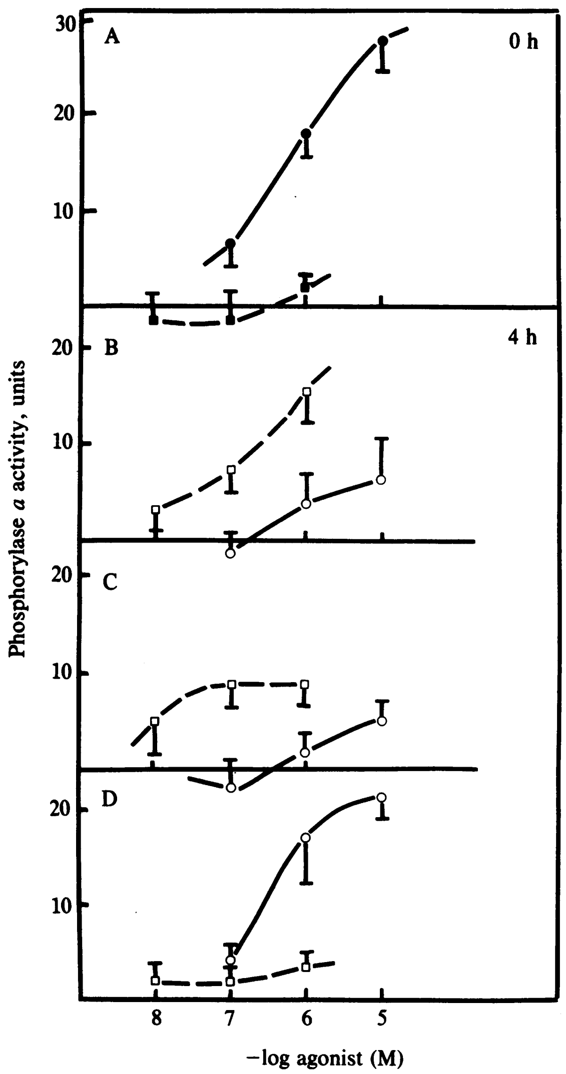

FIG. 1. Effects of regular or fatty-acid-free BSA on the time-dependent conversion from α_1- to β-adrenergic glycogenolysis in isolated rat liver cells. (*A*) Zero-hour control. (*B*) Four-hour control. (*C*) BSA. (*D*) Fatty-acid-free BSA. Activation of phosphorylase by phenylephrine (Phe; circles, solid lines) and isoproterenol (Iso; squares, dashed lines) was assayed in 0-hr (solid symbols) and 4-hr cells (open symbols). Basal phosphorylase *a* activity was 15.7 ± 2.5 units (*A*), 20.0 ± 3.9 units (*B*), 19.9 ± 3.0 units (*C*), and 18.7 ± 3.4 units (*D*). Results represent mean and SEM of seven experiments. The presence of BSA or fatty-acid-free BSA in 0-hr cells (not shown) did not influence the effects of either phenylephrine or isoproterenol: 10 μM phenylephrine increased phosphorylase *a* activity by 27.5 ± 3.6 units (control), 23.4 ± 3.6 units (fatty-acid-free BSA), or 23.5 ± 4.8 units (BSA). The corresponding effects of 1 μM isoproterenol were 1.8 ± 1.4, 3.9 ± 1.7, and 3.4 ± 3.6 units.

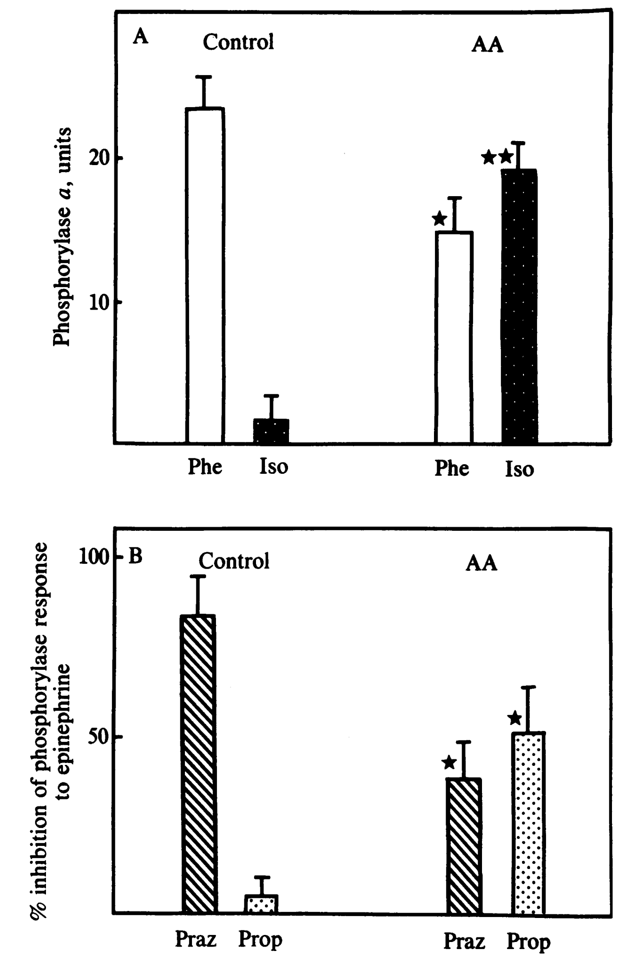

FIG. 2. Effect of arachidonic acid on α_1- and β-receptor mediated activation of glycogen phosphorylase in freshly isolated rat liver cells. (*A*) Maximal response to phenylephrine (Phe, 10 μM) and isoproterenol (Iso, 1 μM), in the absence (control) or presence of arachidonic acid (AA, 10 μg/ml). Cells were exposed to arachidonic acid for 20 min before assaying the effects of the adrenergic agonists. Basal phosphorylase *a* activity was 20.5 ± 2.1 unit in the absence and 28.5 ± 1.9 unit in the presence of arachidonic acid. (*B*) Inhibition of the effect of 1 μM epinephrine by 0.1 μM prazosin (Praz) or 1 μM propranolol (Prop) in control and arachidonic acid-exposed cells. In the absence of antagonists, epinephrine increased the activity of phosphorylase by 31.8 ± 2.4 and 26.3 ± 2.8 units, in the absence and presence of arachidonic acid, respectively. The effect of epinephrine in the presence of the antagonists was calculated as the increase in phosphorylase *a* activity over the baseline in the presence of the antagonist alone. Vertical bars indicate SE, n = 9 (*A*) or 3 (*B*). Stars indicate significant difference from corresponding control values, one star $P < 0.05$, two stars $P < 0.005$.

Table 1. Effect of saturated fatty acids on the adrenergic activation of phosphorylase in freshly isolated rat liver cells

	Increase in phosphorylase *a* activity, units	
	Phenylephrine (10 μM)	Isoproterenol (1 μM)
Control	21.0 ± 3.1	2.7 ± 2.9
Stearic acid*	16.8 ± 3.1	4.1 ± 1.8
Palmitic acid*	16.5 ± 2.8	2.4 ± 3.0

Values are means ± SEM from four separate experiments. Baseline phosphorylase activity was 16.9 ± 1.0 units (control), 30.4 ± 4.7 units (stearic acid), and 28.0 ± 4.4 units (palmitic acid).
*Cells were exposed to the fatty acids (10 μg/ml) or vehicle only for 20 min before assaying the effect of the adrenergic agonists.

that the presence of fatty-acid-free BSA prevents the time-dependent conversion of the adrenoceptor response of hepatocytes strongly suggests that a fatty acid released by the cells plays a key role in the altered receptor response.

The predominant fatty acid contained in the 2 position of membrane phospholipids is arachidonic acid, and it is possible that this is the critical factor removed by the fatty-acid-free BSA. To test this, we examined the effect of exogenous arachidonic acid added to suspensions of freshly isolated liver cells. After the standard 30 min preincubation period, cells were incubated for an additional 20 min with vehicle only (0.1% ethanol) or with arachidonic acid (10 μg/ml). As shown in Fig. 2*A*, arachidonic acid significantly reduced the effect of phenylephrine on phosphorylase, while isoproterenol became a potent activator. This shift from a pure α_1 to a mixed α_1/β type response is also indicated by changes in the effects of receptor antagonists. In control cells, activation of phosphorylase by the mixed α/β agonist epinephrine is blocked by the α_1-antagonist prazosin but not by the β-receptor blocker propranolol. In cells exposed to arachidonic acid, the effect of epinephrine is partially inhibited by either prazosin or propranolol (Fig. 2*B*). When tested in similar experiments, the saturated fatty acids stearic and palmitic acid did not influence the adrenoceptor response (Table 1). Thus, arachidonic acid appears either to directly induce a shift from a pure α_1 to a mixed α/β type response or to accelerate the similar, time-dependent change.

Arachidonic acid is the precursor of eicosanoids, generated predominantly through the cyclooxygenase or lipoxygenase pathways. To test if the effects of arachidonic acid on hepatic adrenoceptors are mediated by a metabolite, freshly isolated hepatocytes were preincubated with the cyclooxygenase inhibitor ibuprofen or the lipoxygenase inhibitor NDGA before the addition of arachidonic acid and subsequent testing of the adrenergic activation of phosphorylase. As illustrated in Fig. 3, the reduction of the effect of phenylephrine and the simultaneous emergence of a response to isoproterenol caused by arachidonic acid were completely prevented by 0.2 μM ibuprofen, while the same concentration of NDGA only slightly attenuated these changes. However, when used at a 10 times higher concentration, both inhibitors were effective in preventing the effects of arachidonic acid (data not shown).

The time-dependent change in the adrenoceptor response was also prevented by ibuprofen (Fig. 4), when used at a slightly higher concentration (1 μM) to offset its gradual uptake and metabolism by the cells (1) during the 4-hr incubation. A similar, although smaller, effect by the same concentration of indomethacin (Fig. 4) strongly suggests that prevention of the change in the adrenoceptor response is due to inhibition of cyclooxygenase.

Inhibitors of protein or mRNA synthesis prevent the conversion of the hepatic adrenoceptor response in primary cultured cells (11) or during *in vitro* incubation of isolated hepatocytes (16). To test whether the more rapid shift in receptor response caused by exogenous arachidonic acid is also dependent on the synthesis of new protein, this effect was also tested in the presence of actinomycin D. In agreement with earlier findings, the presence of actinomycin D (0.25 μg/ml) in the medium during incubation of liver cells for 4 hr prevented the inverse changes in the effects of phenylephrine and isoproterenol (Table 2). However, the same concentration of actinomycin D did not influence the similar, although smaller, changes caused by arachidonic acid (10 μg/ml) in freshly isolated cells (Table 2).

DISCUSSION

We have suggested that increased membrane phospholipase A_2 activity is involved in the conversion from α_1- to β-receptor mediated glycogenolysis that occurs upon *in vitro* incubation of isolated rat liver cells (12, 16). This can now be confirmed and extended. The presence of fatty-acid-free BSA in the incubation medium prevents the time-dependent change in the adrenoceptor response, whereas the same concentration of regular BSA does not. This is interpreted to

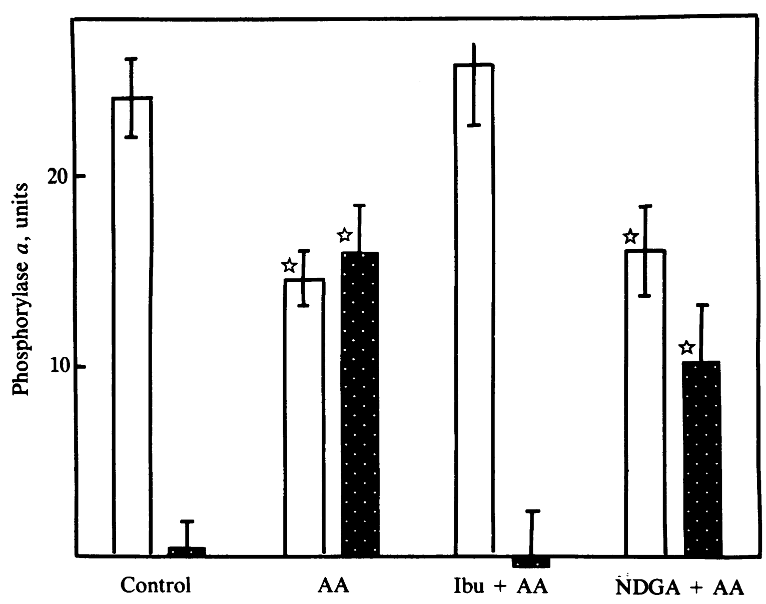

FIG. 3. Effects of ibuprofen (Ibu) and nordihydroguaiaretic acid (NDGA) on the arachidonic acid (AA)-induced change in the adrenergic activation of phosphorylase in freshly isolated rat liver cells. The columns represent the maximal increase in phosphorylase *a* activity caused by phenylephrine (10 μM, open columns) or isoproterenol (1 μM, dotted columns). Vertical bars are SE of the means, obtained in nine (control and AA) or five experiments (Ibu and AA, NDGA and AA). Stars indicate significant difference from corresponding values in controls ($P < 0.05$).

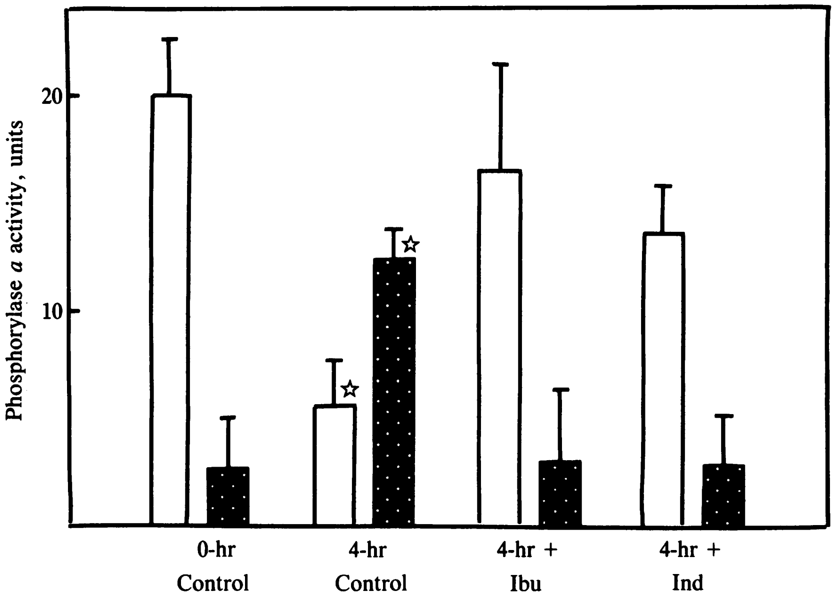

FIG. 4. The effects of ibuprofen and indomethacin on the time-dependent change in the adrenergic activation of phosphorylase in rat liver cells. The columns represent the maximal increase in phosphorylase *a* activity caused by phenylephrine (10 μM, open columns) or isoproterenol (1 μM, dotted columns). Vertical bars are SE, $n = 4$. Ibuprofen (Ibu) or indomethacin (Ind, 1 μM) was added to the cells at 0 hr and again at 2 hr to replace drug lost through uptake and metabolism by the cells. After 4 hr the cells were centrifuged and resuspended in fresh medium containing 1 μM of the inhibitors or vehicle only (4-hr controls) and were assayed with the adrenergic agonists within 30 min. Stars indicate significant difference from corresponding 0-hr values ($P < 0.05$).

indicate that fatty acid(s) generated by liver cells and released into the medium are involved in the suppression of the α_1-receptor-mediated response as well as in the simultaneous emergence of the β-receptor mediated response. The finding that fatty-acid-free BSA does not influence the response pattern of freshly isolated cells suggests that release of the relevant fatty acids is low in these cells but it is increased during their *in vitro* incubation.

When rat liver cells are isolated and placed in primary culture, they show increased growth and proliferation and reciprocally decreased differentiated, liver-specific functions (17). In certain neural cell lines, polyunsaturated fatty acids promote multiplication (18) and inhibit the differentiated response of neurite extension (19). Therefore, increased release of certain fatty acids may have a role in the dedifferentiation that occurs upon loss of cell-to-cell contact. Since dedifferentiation is a common denominator among conditions in which a conversion from α_1- to β-receptor mediated glycogenolysis occurs in the rat liver (1), a possible role of fatty acids in the altered receptor response is plausible. Unsaturated fatty acids have been also implicated in the

Table 2. Effects of actinomycin D on the conversion of α- to β-adrenergic receptor-mediated glycogenolysis in isolated rat liver cells induced by arachidonic acid or by incubation of cells for 4 hr

		Increase in phosphorylase *a* activity, units		
Cells	Treatment	Phenylephrine (10 μM)	Experiment, no.	Isoproterenol (1 μM)
0-hr	None	29.2 ± 2.5	5	2.3 ± 2.0
4-hr	None	8.5 ± 2.0*	5	18.6 ± 4.8*
0-hr	Actinomycin D	26.9 ± 3.3	4	5.1 ± 3.4
4-hr	Actinomycin D	31.0 ± 6.2	4	6.8 ± 3.2
0-hr	Arachidonic acid	16.4 ± 2.8**	5	12.7 ± 2.2**
0-hr	Actinomycin D + arachidonic acid	20.5 ± 2.1**	5	17.4 ± 5.2**

The concentration of actinomycin D and arachidonic acid was 0.25 μg/ml and 10 μg/ml, respectively. Incubation of 0-hr cells with arachidonic acid was for 20 min, and actinomycin D was added to the medium 10 min before arachidonic acid. Values are means ± SEM from either four or five experiments. Asterisks indicate significant difference from values in untreated 0-hr cells, $*P < 0.005$, $**P < 0.05$.

regulation of other, receptor mediated processes: Wilkening and Nirenberg (20) have documented that linoleic acid, a precursor of arachidonic acid, is required for the full expression of the long-lived morphine-dependent increase in adenylate cyclase activity in a neuroblastoma–glioma hybrid cell line (20).

The predominant fatty acid in the 2 position of membrane phosphatides is arachidonic acid. The experiments with exogenous fatty acids strongly support the role of arachidonic acid in the conversion of the hepatic adrenoceptor response. The rapid change in the adrenergic response pattern caused by arachidonic acid is similar, although less complete than the change that develops upon incubation of the cells for 4 hr. The specificity of the effects of arachidonic acid is indicated by the lack of similar effects of palmitic and stearic acids. Thus, arachidonic acid either mimics or accelerates the effects of prolonged *in vitro* incubation.

The findings with inhibitors of arachidonic acid metabolism further suggest that an oxidation product rather than arachidonic acid itself is responsible for the observed change in receptor response. At the low concentrations used, ibuprofen is a relatively selective inhibitor of cyclooxygenase, and its reversal of the effects of arachidonic acid suggests that these effects are mediated by a cyclooxygenase product. The lack of a similar inhibition by the same low concentration of NDGA is compatible with this possibility, although at a higher concentration NDGA was also an effective inhibitor. Recent studies in bovine retina indicate that although NDGA is a highly potent and selective inhibitor of lipoxygenase, at micromolar concentrations it can also inhibit cyclooxygenase (21). The time-dependent conversion of the adrenoceptor response was also inhibited by both ibuprofen and indomethacin, which suggests that a cyclooxygenase product is involved in the endogenously triggered change. A similar effect was earlier reported with 10 μM indomethacin, at which concentration phospholipase A2 activity in purified liver plasma membranes was also inhibited (16). However, as the latter effect is only present at concentrations of 3 μM or higher (22), these results tend to dissociate the two effects and suggest that inhibition of the change in receptor response by indomethacin is related to inhibition of cyclooxygenase. It is interesting to note that in the isolated rat uterus prostaglandin $F_{2\alpha}$, a cyclooxygenase product, was reported to potentiate β-adrenergic relaxation and inhibit α-adrenergic contraction (23).

The time-dependent change in the glycogenolytic response is selective for adrenergic agents (11, 12), and it is not associated, at least initially, with changes in the density of α_1- to β-receptor binding sites or their affinity for antagonist radioligands (12). This narrows the possible site of change to a step immediately following receptor activation but preceding the generation of second messenger signals, which is most likely the coupling of receptors to their postreceptor pathways. In several recent studies, primary culturing of rat liver cells increased the density of β-receptors (24–27) and caused a parallel decrease in the density of α_1-receptors (26, 27). However, these changes take more than 3 hr to appear (27) and 1–2 days to fully develop (26), which suggests that they may be the consequence rather than the cause of the much faster changes in the receptor response. This would be compatible with the suggestion that changes in receptor coupling mechanisms may trigger more delayed changes in the expression of receptors on the cell surface (28).

Actinomycin D inhibits the time-dependent but not the arachidonate-induced conversion of the hepatic adrenoceptor response (Table 2). This indicates that a protein factor(s) is involved in the reciprocal expression of α- and β-adrenergic functions, and that this factor acts at a site proximal to the release of fatty acids and the generation of a regulatory arachidonate metabolite. A plausible target for the regulatory action of this protein could be the enzymatic mechanism involved in the release of arachidonic acid. The mechanism of action of the arachidonate metabolite is not yet clear. The reciprocal nature of the changes in α_1- and β-receptor activity suggests that a common mechanism is involved in the inverse coupling of the two receptors to their respective postreceptor pathways (12). Recently, Itoh *et al.* (29) reported that conversion from α_1- to β-receptor-mediated glycogenolysis during primary culturing of rat liver cells is accompanied by an increase in the ADP-ribosylation of the inhibitory guanine nucleotide-binding protein N_i and that inhibitors of ADP-ribosyltransferases delayed both changes. This has led to the proposal that, in the rat liver, N_i may be linked to β-adrenergic receptors in an inhibitory way (29). Since N_i has also been implicated in receptor-mediated calcium gating (30), it is possible that its inactivation through ADP-ribosylation can result in simultaneous activation of β- and inhibition of α_1-receptor-mediated functions in the rat liver. One might speculate that an arachidonate metabolite could control the functional state of a relevant N protein, possibly by regulating the activity of an endogenous ADP-ribosyltransferase. However, a role for N_i in the hepatic α_1-adrenergic response has not yet been demonstrated directly. Furthermore, the selectivity of the time-dependent conversion for adrenergic agents is difficult to reconcile with changes in N protein function without additional, receptor-specific changes. Clarification of the nature of such changes must await further studies.

This work was supported by a grant from the Medical Research Council of Canada. Fellowship support to E.J.N.I. was provided by the Canadian Heart Foundation.

1. Kunos, G. (1984) *Trends Pharmacol. Sci.* **5**, 380–383.
2. Studer, R. K. & Borle, A. B. (1982) *J. Biol. Chem.* **257**, 7987–7993.
3. Sherline, P., Eisen, H. & Glinsmann, W. (1974) *Endocrinology* **94**, 935–939.
4. Preiksaitis, H. G. & Kunos, G. (1979) *Life Sci.* **24**, 35–42.
5. Malbon, C. C., Li, S. & Fain, J. N. (1978) *J. Biol. Chem.* **253**, 8820–8825.
6. Chan, T. M., Blackmore, P. F., Steiner, K. E. & Exton, J. H. (1978) *J. Biol. Chem.* **254**, 2428–2433.
7. Christoffersen, T. & Berg, T. (1975) *Biochim. Biophys. Acta* **381**, 72–77.
8. Aggerbeck, M., Ferry, N., Zafrani, E. S., Billon, M. C., Barouki, R. & Hanoune, J. (1983) *J. Clin. Invest.* **71**, 476–486.
9. El-Refai, M. F. & Chan, T. M. (1982) *FEBS Lett.* **146**, 397–402.
10. Huerta-Bahena, J., Villalobos-Molina, R., Corvera, S. & Garcia-Sainz, J. A. (1983) *Biochim. Biophys. Acta* **763**, 112–119.
11. Okajima, F. & Ui, M. (1982) *Arch. Biochem. Biophys.* **213**, 658–668.
12. Kunos, G., Hirata, F., Ishac, E. J. N. & Tchakarov, L. (1984) *Proc. Natl. Acad. Sci. USA* **81**, 6178–6182.
13. Hirata, F. (1981) *J. Biol. Chem.* **256**, 7730–7733.
14. Preiksaitis, H. G., Kan, W. H. & Kunos, G. (1982) *J. Biol. Chem.* **257**, 4321–4327.
15. Lowry, O. H., Rosebrough, N. J., Farr, A. L. & Randall, R. J. (1951) *J. Biol. Chem.* **193**, 265–275.
16. Kunos, G. & Ishac, E. J. N. (1985) *J. Cardiovasc. Pharmacol.* **7**, Suppl. 6, 87–92.
17. Nakamura, T., Nakayama, Y., Teramoto, H., Nawa, K. & Ichihara, A. (1984) *Proc. Natl. Acad. Sci. USA* **81**, 6398–6402.
18. Yavin, E., Yavin, Z. & Menkes, J. (1975) *Neurobiology* **5**, 214–220.
19. Monard, D., Rentsch, M., Schuerch-Rathgeb, Y. & Lindsay, R. M. (1977) *Proc. Natl. Acad. Sci. USA* **74**, 3893–3897.
20. Wilkening, D. & Nirenberg, M. (1980) *J. Neurochem.* **34**, 321–326.
21. Birkle, D. L. & Bazan, N. G. (1984) *Prostaglandins* **27**, 203–216.
22. Kunos, G., Tchakarov, L., Ishac, E. J. N. & Kan, W. H. (1985) in *Adrenergic Receptors: Molecular Properties and Therapeutic Implications*, Symposia Medica Hoechst, eds. Lefkowitz, R. J. & Lindenlaub, E. (Schattauer, Stuttgart), Vol. 20, in press.
23. Chaud, M., Borda, E., Franchi, A. M., Sterin-Borda, L., Gimeno, M. F. & Gimeno, A. L. (1982) *Eur. J. Pharmacol.* **78**, 449–456.
24. Nakamura, T., Tomomura, A., Noda, C., Shimoji, M. & Ichihara, A. (1983) *J. Biol. Chem.* **258**, 9283–9289.
25. Refsnes, M., Sanders, D., Melien, O., Sand, T. E., Jacobsen, S. & Christoffersen, T. (1983) *FEBS Lett.* **164**, 291–298.
26. Nakamura, T., Tomomura, A., Kato, S., Noda, C. & Ichihara, A. (1984) *J. Biochem. (Tokyo)* **96**, 127–136.
27. Schwartz, K. R., Lanier, S. M., Carter, E. A., Homcy, C. J. & Graham, R. M. (1985) *Mol. Pharmacol.* **27**, 200–209.
28. Mahan, L. C., Koachman, A. M. & Insel, P. A. (1985) *Proc. Natl. Acad. Sci. USA* **82**, 129–133.
29. Itoh, H., Okajima, F. & Ui, M. (1984) *J. Biol. Chem.* **259**, 15464–15473.
30. Okajima, F. & Ui, M. (1984) *J. Biol. Chem.* **259**, 13863–13871.